Gender-based temporal dynamics in sexual behaviour in Britain

Summary

Main research question

Is gender a significant factor driving change in the number of heterosexual partners reported by the British populace between 2000 and 2010?

Main findings

- Men report significantly higher numbers of heterosexual partners than do women
- There is a small but non-significant change in sexual activity over time
- Gender does not significantly drive changes in the number of heterosexual partners over time
- There is a time-invariant gender gap in sexual activity for the British populace

Introduction

Context

Despite a century of epidemiological research and maturing public health policy, sexual wellbeing and disease prevention remain a chief concern in the United Kingdom (Mohammed et al., 2018). Gender is known to shape sexual lifestyle, being associated with attitudes towards sexual activity (Kurpisz et al., 2016), sexual vulnerability (Ricardo et al., 2006), and the frequency of engagement in risky sexual behaviour (Olapegba et al., 2013). While the manifestation of these effects varies globally (Wellings et al., 2006), it is evident that gendered differences in sexual behaviour are often modulated by perceived sociocultural roles (Cubbins & Tanfer, 2000; Ricardo et al., 2006). It is, therefore, likely that the influence of gender on sexual activity changes as social norms evolve over time. This report examines whether gender can explain temporal dynamics in the sexual behaviour of the British populace. Clarifying this relationship will provide critical insights into the drivers of STI transmission patterns, inform demographically-structured health intervention strategies, and guide campaigns that aim to refine public sex education.

The analyses that aim to address this question are based on data from *British National Surveys of Sexual Attitudes and Lifestyles* (NSSAL) surveys, which have aimed to catalogue sexual attitudes in Britain by capturing answers to diverse questions pertaining to sexual education, experiences, and lifestyle. We gauge time-related changes in sexual activity (represented by numbers of heterosexual partners) and their link to gender by contrasting two surveys, NSSAL-2 (conducted: 1999-2001) and NSSAL-3 (conducted: 2010-2012), that span the first decade of the 21st century.

Conceptual model and variables of interest

We propose a base model which relates sexual activity in the British populace to the gender of individuals and the decade in which they are interviewed:

```
sexual activity ~ gender + time + gender*time
```

Our hypotheses are focused on the evaluation of the interaction term in the above model. The null hypothesis states that there is no difference between men and women in how their number of heterosexual partners changes over time. The alternative hypothesis states that gender significantly modulates the change in heterosexual partners over time.

From the NSSAL survey data, we opt to use the reported number of heterosexual partners in the past year (**het1yr**; count) as a dependent variable, as it provides an estimate of sexual activity most reflective of respondent sexual lifestyle close to the time of interview. Gender (**rsex**; binary; 0=female, 1=male) and the decade of interview (**reportingperiod**; binary, 0=2000, 1=2010) are used as categorical predictors. Respondent age at the time of interview (**dage**; continuous) is considered as a covariate given that age is frequently correlated with sexual activity (Kontula, 2009; Hawton et al., 1994), and that NSSAL-3 placed specific emphasis on sampling younger individuals.

Analysis

Data preparation

Prior to analysis, data from NSSAL-2 and NSSAL-3 were combined and filtered to variables of interest. The NSSAL-3 dataset was first trimmed to retain only respondents with ages <= 44 to match the age group sampling distribution for NSSAL-2. Columns pertaining to respondent serial identifiers, primary sampling units (PSUs), and survey strata were matched between the datasets by inspection of their respective code books. Responses in the dataset that were uncertain (e.g., 1 or more heterosexual partners) or non-answers (e.g., not applicable) for the dependent variable were removed. Thereafter, only records with complete information across all columns in the dataset were kept. This filtering procedure led to a final dataset consisting of 20466 observations, maintaining 75% of the original 27272 observations across NSSAL-2 and NSSAL-3.

Descriptive statistics

Descriptive statistics were calculated to summarize and compare the distributions of the dependent variable in each of the four cells formed by combinations of the two main predictors, gender and time. Typically, the complex structure of the survey data at hand must be accounted for wherever possible using survey weights, PSUs, and strata to ensure that data appropriately capture population-level patterns. This can be done using the `survey` R package. However, we report descriptive statistics and results for normality and homogeneity of variance tests based on raw count data,a s methods for several required operations are not packaged with `survey`. The survey design is, however, appropriately accounted for when conducting model fits.

Values of the mean (M), variance (V), and skewness (Sk) were used to characterize the central tendency and spread of data. The cell concerning males from 2010 showed the

highest overall number of heterosexual partners in the last year (M=1.69; V=9.86; Sk=8.40), followed by males from 2000 (M=1.62; V=7.53; Sk=9.85), females from 2010 (M=1.37; V=6.80; Sk=34.6), and females from 2000 (M=1.20; V=1.89; Sk=11.3). For all cells, the distribution of the dependent variable was highly positively skewed based on skewness estimates and visual inspection of boxplots (Figure 1). Quantile-quantile plots indicated severe deviations from normality for all cells, likely accounted for by the extreme skewness of the data.

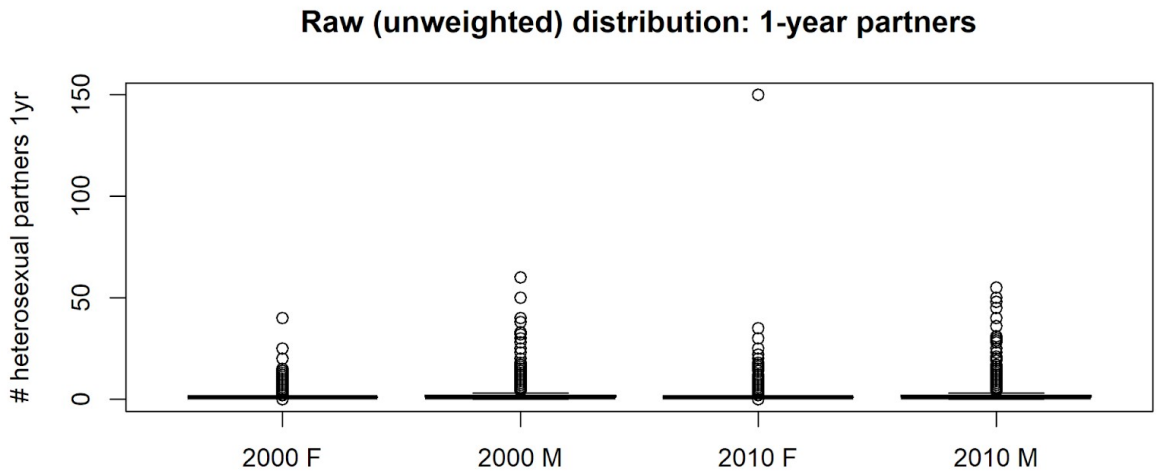

Figure 1. Boxplots of raw survey count data for the dependent variable het1yr by factorial cell. Number in group name refers to the survey year; F = female; M = male.

Homogeneity of variance was analyzed across the groups using Levene's test. Levene's test statistic was large (63.897), with a p-value of less than 0.001, providing strong evidence against equal variances between groups.

Model selection

The critical modelling assumption of independence of observations appears to hold for the data, given that respondent serial identification numbers are unique throughout the dataset. However, preliminary analysis indicated the assumptions required for traditional linear regression, and analysis of variance (ANOVA) - e.g., normality and homoscedasticity of residuals - could not be satisfied. Accordingly, we opted to use generalized linear models (GLMs) to address the research question, as they: (1) allow for the analysis of non-normally distributed data, (2) can explicitly model non-constant variances, and (3) provide a means to avoid direct transformation of the dependent variable through link functions.

Poisson regression with the standard log-link was initially considered due to the outcome variable being low-valued non-negative integers (i.e. counts). However, inspection of Poisson model fits revealed substantial overdispersion (overdispersion ratio = 3.828; based on model deviance and residual degrees of freedom), violating a key assumption of the model that the mean is equal to variance.

To account for overdispersion, and the shape of count data, a quasi-Poisson model was employed. Due to the complex survey design of the NSSAL data, models were run using the `svyglm` function from the `survey` R package to incorporate survey weights, clustering, and stratification in order to produce estimates and standard errors which correctly reflect the population parameters despite the survey sampling design.

In implementing the GLM approach, we constructed a naive model that represents our initial conceptual model (`sexual activity ~ gender + time + gender*time`). To account for potential confounding effects of age on the dependent variable, and the bias in age group sampling between datasets, we also explore a more sophisticated model that includes respondent age as an additive predictor (`sexual activity ~ gender + time + gender*time + age`). The selection of this variable as a covariate is meaningful as it is not accounted for in the weights provided in the NSSAL data sets. We compare the fit of the two models using the ANOVA method with a chi-square test.

Inference

Comparing the two defined GLMs, there is significant evidence (Likelihood ratio test statistic=54.644; Chi-square $p<0.001$) in favour of the more sophisticated model containing respondent age as a covariate. In this model, age is significantly associated with respondent age (IRR = 0.983; $p < .001$). We, therefore, find it valuable to retain it as a working model, and focus on interpretation of this model and its other coefficients (Figure 2) conditional on respondent age. Doing so provides a basis for several key inferences regarding general sexual behaviour trends in Britain between 2000 and 2010. In the below results summary, incidence risk ratios (the exponential form of the model coefficient) are reported, and can be interpreted as the proportional change expected in the dependent variable given a unit change in the independent variables.

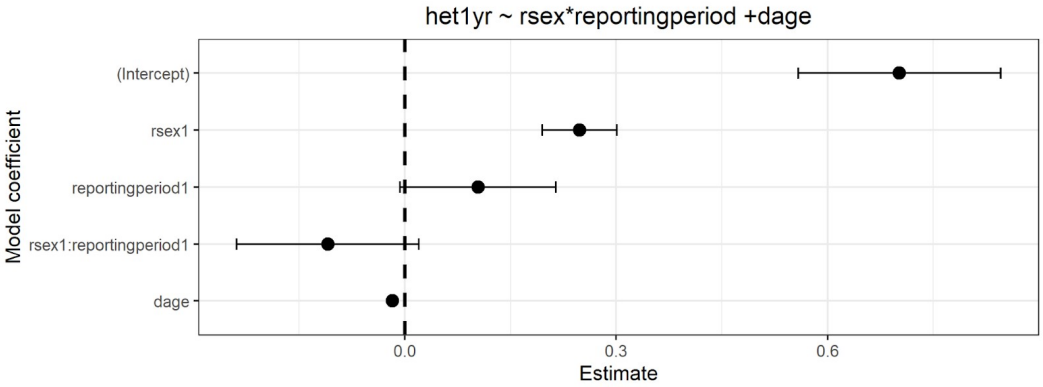

Figure 2. Plot of the coefficients and their estimates based on the “sophisticated” GLM containing the predictors of interest and age as a covariate. Means are represented as points, with bars indicating the 95% confidence intervals. Values shown are raw coefficients (i.e., not exponentiated to form IRR values).

Gender-Based Disparities

There is strong evidence to infer that gender remains a primary determinant of reported sexual partner counts. At the baseline, men reported an incidence rate of heterosexual partners approximately **28% higher** than women (IRR = 1.282; $p < .001$), a difference that is highly statistically significant.

Temporal Shifts in Behaviour

We can infer that there is no strong shift in reporting or behaviour over the decade for the reference group (females). While this group saw an estimated **11% increase** in their annual partner rates, this change was **not significant** (IRR = 1.109; $p = 0.067$).

Stability of the Gender Gap

Regarding the primary research question, the model does **not provide sufficient evidence** to infer that gender significantly moderated the change in partner counts over time. The interaction effect (IRR = 0.896, $p = 0.097$) suggests that while the "gender gap" appeared to narrow in the sample in 2010—with women’s rates rising while men’s remained stable—this trend cannot be confidently generalized to the broader British population at the 95% confidence level.

Visualized effects

The above interpretations are represented well in an interaction plot (Figure 3). The gender gap in the number of heterosexual partners reported between genders appears large at both reporting periods. While there is a slight increase in partners observed for

women, a change in number of partners reported is virtually absent in men, likely accounting for the lack of overall effect of reporting period on the dependent variable.

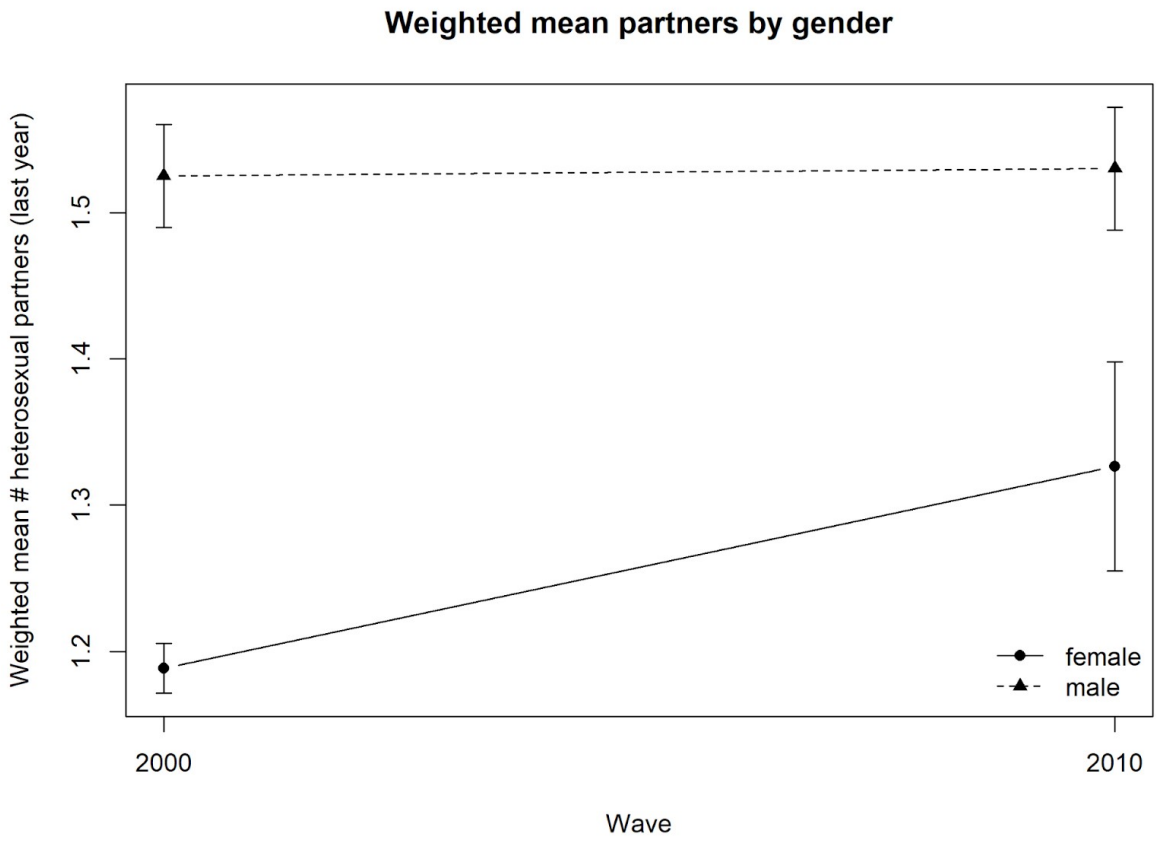

Figure 3. Interaction plot of means and standard errors for het1yr in each factorial group under study.

Reliability of Estimates

Assessing deviance in the fitted quasi-Poisson model incorporating survey weights, we observe high overdispersion (dispersion parameter = 4.32), validating the choice of a quasi-Poisson model over the base Poisson model. Notably, this approach produces more conservative inference relative to a standard Poisson specification. Diagnostic plots (Figure 4) suggest that estimates are not highly driven by influential observations, with only one data point having a high Cook's distance. However, this value still falls below the typical threshold of 1 used to identify highly influential data.

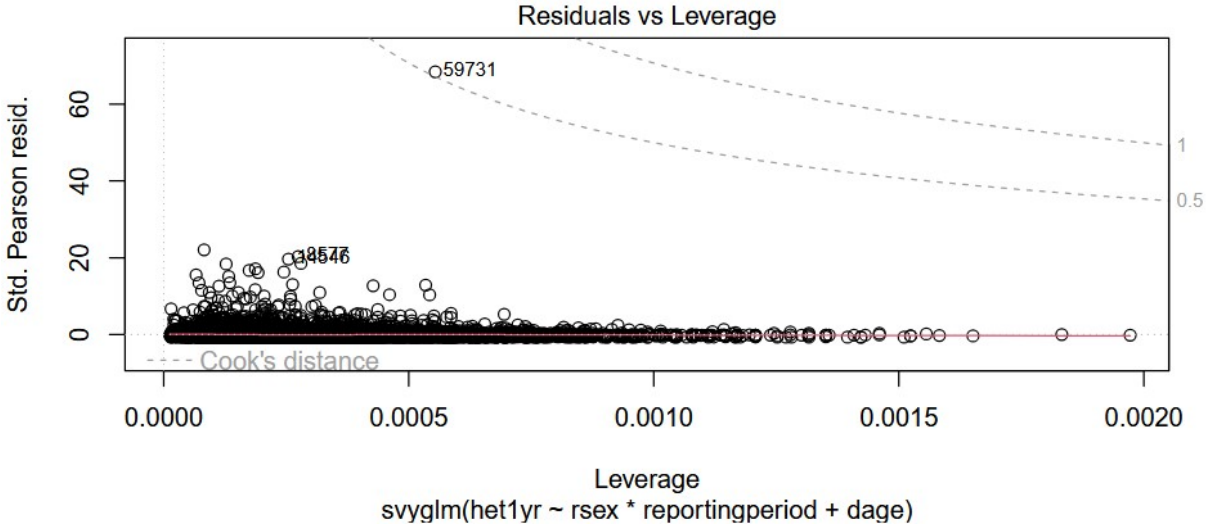

Figure 4. Plot of data residuals versus leverage, with Cook's distance thresholds of 0.5 and 1 plotted.

Limitations

Survey data is susceptible to recall bias (Wellings et al., 2001) and is thus an inherent limitation for this analysis. Considering the number of heterosexual partners in the past 1 year as opposed to in the past 5 years should alleviate some of this bias as participants are likely to have better recollection over the shorter time period.

The report specifically aims to address whether gender can explain the temporal shifts in the sexual behaviour of the British populace. Therefore, it was important for us to include a time variable into our models to determine whether there was a significant change in trend over time. As a result, we combined the NSSAL-2 and NSSAL-3 datasets which brought to the fore the challenge of the datasets having different weightings and clusterings. Access to weighting and clustering variables that accounted for these was restricted. We accounted for these survey design elements for both datasets; however, we used now-outmoded strata and weightings variables for the 2000 survey data. Therefore, a limitation to our analysis is that the findings reported for the 2000 period are likely to be more sample specific and susceptible to Type I error (Bell et al., 2012).

In our analysis we did not consider the effect of additional "third variables" (i.e., confounders) beyond respondent age. Thus, there is potential that our results do not account for the effect of these variables. We recommend that further analysis be conducted to examine the effect of potentially important covariates, such as relationship status, culture, sexual education, age of first sexual contact etc.

Conclusion

This analysis investigated whether gender was a significant factor driving change in the number of heterosexual partners reported by the British populace between 2000 and 2010. We found strong evidence that gender is associated with the number of heterosexual partners reported, with men consistently reporting higher counts than women. Contrary to the expectation of a change in behaviours over time, we find little evidence of a strictly time-related increase in heterosexual partnerships between 2000 and 2010. Additionally, the non-significant interaction between gender and survey period indicates that gender does not significantly influence the change in partner numbers over time. These findings suggest that while gender differences persist, behavioural trends over the decade were broadly similar for men and women.

The findings from this analysis provide insight into the efficacy of the sexual health services that were implemented in the wake of reports and population analyses based on the seminal NSSAL-1 survey data. Although these services have proven effective in reducing promiscuity among the British populace, this decline is likely not attributed to gender specific initiatives. While the lack of a significant interaction suggests that gender did not drive the *rate of change* of sexual activity between 2000 and 2010, the persistent and significant disparity in reported partners indicates that gender remains a critical demographic lens for sexual health interventions.

References

Bell, B.A., Onwuegbuzie, A.J., Ferron, J.M., Jiao, Q.G., Hibbard, S.T. Kromrey, JD., 2012. Use of Design Effects and Sample Weights in Complex Adolescent Health Surveys. *American Journal of Public Health, 102(7), pp 1399–1405*

Cubbins, L.A. and Tanfer, K., 2000. The influence of gender on sex: A study of men's and women's self-reported high-risk sex behavior. *Archives of Sexual Behavior*, *29*(3), pp.229-257.

Hawton, K., Gath, D. and Day, A., 1994. Sexual function in a community sample of middle-aged women with partners: effects of age, marital, socioeconomic, psychiatric, gynecological, and menopausal factors. *Archives of sexual behavior*, *23*(4), pp.375-395.

Kontula, O. and Haavio-Mannila, E., 2009. The impact of aging on human sexual activity and sexual desire. *Journal of sex research*, *46*(1), pp.46-56.

Kurpisz, J., Mak, M., Lew-Starowicz, M., Nowosielski, K., Bieńkowski, P., Kowalczyk, R., Misiak, B., Frydecka, D. and Samochowiec, J., 2016. Personality traits, gender roles and sexual behaviours of young adult males. *Annals of general psychiatry*, *15*(1), p.28.

Mohammed, H., Blomquist, P., Ogaz, D., Duffell, S., Furegato, M., Checchi, M., Irvine, N., Wallace, L.A., Thomas, D.R., Nardone, A. and Dunbar, J.K., 2018. 100 years of STIs in the UK: a review of national surveillance data. *Sexually transmitted infections*, *94*(8), pp.553-558.

Olapegba, P.O., Idemudia, E.S. and Onuoha, U.C., 2013. Gender differences in responsible sexual behaviour of in-school adolescents. *Gender and Behaviour*, *11*(1), pp.5316-5322.

Ricardo, C., Barker, G., Pulerwitz, J. and Rocha, V., 2006. Gender, sexual behaviour and vulnerability among young people. In *Promoting young people's sexual health* (pp. 61-78). Routledge.

Wellings, K., Collumbien, M., Slaymaker, E., Singh, S., Hodges, Z., Patel, D. and Bajos, N., 2006. Sexual behaviour in context: a global perspective. *The Lancet*, *368*(9548), pp.1706-1728.